BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 2
Duration: 25 minutes

Semester: Spring 2025
Full Marks: 15

CSE 340: Computer Architecture

Name: Solution	ID:	Section: 01

1. if (a > b or b < c):
  a = b
elif (a == 5):
  c = 10
else:
  d = d + 10

Construct the equivalent RISC-V code of the above high-level code. Variables a,b,c,d are associated with register X_{11}, X_{12}, X_{13}, X_{14} respectively [4]

Answer:

Blt x12, x11, Cond1
Blt x12, x13, Cond1
Addi x10, x0, 5
~~Beq~~ Beq x11, x10, Elif
Addi x14, x14, 10
Beq x0, x0, Exit

Cond1:
Add x11, x12, x0
Beq x0, x0, Exit
Elif:
Addi x13, x0, 10
Exit:

2. In the RISC-V architecture, the size of each register is 64 bits.
Suppose we are working with a prototype version of RISC-V where the size of each register is 32 bits. For this prototype version, what would be the required size of the immediate field in case of SRLI instruction and explain why? [2+1]

Answer: Register size = 32 bits. So, valid shifting 1 to 31.

Maximum shifting 31 times.

$\underbrace{11111}$ → 5 bits.

Size of immediate field = 5 bits.

3. Find the RISC-V code of the following machine Codes.

i. 0x**00AA9B93** **[3]**

Answer: $\underbrace{0000\ 000}\ \underbrace{0\ 1010}_{imm}\ \underbrace{1010\ 1}_{rs1}\underbrace{001}_{f3}\ \underbrace{1011\ 1}_{rd}\underbrace{001\ 0011}_{opcode}$ → I tpe

↳ to find the exact ins.

Slli x23, x21, 10

ii. 0x**08C2BAB3** **[3]**

Answer: $\underbrace{0000\ 100}_{f7}\underbrace{0\ 1100}_{rs2}\ \underbrace{0010\ 1}_{rs1}\underbrace{011}_{f3}\ \underbrace{1010\ 1}_{rd}\underbrace{011\ 0011}_{opcode}$ → R type

OR x21, x5, x12

combination of f3 & f7 to find the exact ins.

4. In the RISC-V architecture, what is the reason behind splitting the immediate field in S-type instruction? [2]

Answer: Because, we want to maintain the instruction formats same as much as we can.

In S type → to keep rs2 & rs1 fields side by side we divide the immediate into two portions.

Format	Instruction	Opcode	Funct3	Funct7/Funct6
R	ADD	0110011	001	0000000
	SUB	0110011	110	0000001
	AND	0110011	010	0000001
	OR	0110011	011	0000100
	SLL	0110011	101	0000100
I	LD	0000011	000	N/A
	LW	0000011	010	N/A
	LH	0000011	011	N/A
	LB	0000011	001	N/A
	ADDI	0010011	000	N/A
	SLLI	0010011	001	000000
	SRLI	0010011	010	000000
	ORI	0010011	011	N/A
S	SD	0100011	000	N/A
	SW	0100011	001	N/A
	SH	0100011	010	N/A
	SB	0100011	011	N/A